SAMY DERRADJI

Electronic Engineer — Embedded Systems & Hardware Integration

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Profile

Electronic and embedded systems engineer with 3+ years of hands-on experience taking hardware products from concept to production — PCB design, board bring-up, firmware development, and field debugging. Comfortable working across the full hardware lifecycle, from schematic to verified production board. Currently also active in FPGA/ASIC research (RISC-V SoC tapeout) as an added technical edge.

Technical Skills

Microcontrollers: STM32 series, Espressif ESP32 Series, AVR

Embedded Firmware: C/C++, Embedded C, MicroPython, ESP-IDF, STM32Cube

Electronics & PCB: Schematic capture, 2-layer PCB layout (KiCad, EasyEDA, Altium), BOM management, SMT Soldering, board assembly & bring-up

Communication / Wireless: UART, I²C, SPI, CAN, RS485, USB; Wi-Fi, BLE, ESP-NOW

Test & Debug: Oscilloscope, multimeter, logic analyzer, hardware probing, fault analysis, product verification & QA

Digital Design (additional): VHDL, Verilog, FPGA (Xilinx Zynq-7000), basic CPU architecture

Experience

Doctoral Researcher — Low-Power CPU/SoC Design (FPGA/ASIC) *2025 — Present*
Univ. of Blida 1 / CDTA

- Core contributor to QoSoC (low-power RISC-V SoC, fabricated on IHP SG13G2 130nm PDK); led FPGA-level functional verification (SystemVerilog) and RTL/Wishbone integration debug.

Embedded Systems Engineer (Hardware & Firmware) *Sept 2023 — Present*
Techmology — Algiers, Algeria

- Designed and shipped educational robotics / IoT hardware products end-to-end: schematic capture, PCB layout, component selection/BOM, and embedded firmware.
- Developed firmware for STM32 and other MCUs covering motor control, sensor acquisition, and communication interfaces (UART, I²C, SPI, CAN).
- Performed PCB assembly, board bring-up, and hardware debugging using lab equipment (oscilloscope, multimeter, logic analyzer).
- Carried out product testing and verification across multiple finished boards, from prototype to production.

Embedded Systems Intern *August 2022*
Techmology — Algiers, Algeria

- Contributed to a smart IoT greenhouse system, from sensor data capture to visualization.
- Developed ESP32 firmware applying low-power techniques.

Selected Projects

STM32-Based 4-Wheeled Robot Motor Control Board

- Designed and built a compact 2-layer PCB around an STM32F103 MCU to control four motors independently using hall-encoder feedback.
- Implemented firmware in Embedded C for motor control, encoder processing, and communication via UART, CAN, and USB.

Soil Quality Measurement System

- Designed a 2-layer PCB integrating an Atmega328P, SIM800L GSM module, and RS485 soil sensor for remote environmental monitoring.
- Optimized power delivery and signal integrity for stable long-distance data acquisition.

Education

PhD, Microelectronics / Low-Power SoC Design (in progress) — *Univ. of Blida 1 / CDTA* 2025 — Present
Master's Degree, Embedded Systems Electronics — *USTHB* 2021 — 2023
Bachelor's Degree in Electronics — *USTHB* 2018 — 2021

Languages

Arabic (native) | French (fluent) | English (professional fluency)